

Literature Review & Requirements

Gathering: Smart Vision System

1. Literature Review

1.1 Traditional CCTV Monitoring vs. AI-Driven Computer Vision Systems

The evolution of physical security has marked a distinct paradigm shift from passive recording to active, intelligent analysis. Traditional Closed-Circuit Television (CCTV) systems rely fundamentally on human operators for incident detection, inherently suffering from vigilance decrement, cognitive fatigue, and an inability to scale across multiple simultaneous camera feeds. Forensic investigation in traditional systems is strictly reactive, requiring manual review of extensive footage post-incident. Conversely, AI-Driven Computer Vision systems transform cameras into proactive sensory nodes. By applying deep learning models at the edge or cloud level, these systems autonomously extract high-level semantic data (e.g., "person," "weapon," "loitering") from raw pixel arrays. This transition enables real-time automated alerting, predictive operational analytics, and a shift from reactive security to proactive threat mitigation and business intelligence gathering.

1.2 Research Summary: Object Detection Architectures for Retail Environments

The selection of an object detection architecture is critical in retail environments where high pedestrian density, occlusions, and variable lighting are prevalent.

- **Faster R-CNN:** While highly accurate due to its two-stage Region Proposal Network (RPN) architecture, Faster R-CNN operates at approximately 5-7 Frames Per Second (FPS) on standard hardware, rendering it inadequate for real-time, multi-camera SaaS applications.
- **SSD (Single Shot MultiBox Detector):** SSD improves inference speeds significantly through a single-shot approach but often compromises on small-object detection accuracy—a critical failure point in retail aisles where individuals may be partially obscured by shelving.

- **YOLOv8 (You Only Look Once):** As a state-of-the-art, anchor-free object detection model, YOLOv8 provides an optimal equilibrium between inference speed and mean Average Precision (mAP). Its optimized architecture allows for high-throughput processing (>30-60 FPS on contemporary GPUs) while maintaining superior bounding box regression for small to medium objects. For a SaaS platform targeting small businesses, YOLOv8’s balance of low computational overhead and high accuracy ensures viable cloud deployment costs without sacrificing detection reliability.

1.3 State-of-the-Art: Multi-Object Tracking (MOT) via ByteTrack

Detecting objects is only the foundational step; assigning persistent identities across frames is essential for foot traffic counting and heatmap generation. Traditional tracking algorithms (e.g., SORT) discard low-confidence detections, leading to severe Identity Switches (IDSW) in crowded or occluded retail scenarios. More complex approaches like DeepSORT utilize heavy Re-Identification (Re-ID) neural networks, exponentially increasing computational requirements and latency. ByteTrack represents a paradigm shift in Multi-Object Tracking by exploiting high-score detections in the first association stage and strategically recovering low-score detections in a second stage. This approach significantly reduces IDSW in dense environments without the computational bottleneck of feature extraction networks, making it the ideal tracking backbone for a cost-effective, real-time retail analytics SaaS.

2. Stakeholder Analysis

Stakeholder Category	Role / Title	Primary Interests & Goals	Key Pain Points Addressed
Small Business Owners	Proprietor / Investor	Maximize Return on Investment (ROI), ensure asset protection, gain high-level operational insights, and maintain strict compliance with data privacy regulations.	High costs of enterprise-grade security systems; inability to measure physical store performance against digital KPIs; liability risks associated with biometric data storage.
Store Managers	Operations Manager	Optimize staff scheduling based on peak hours, improve store layout, reduce	Reliance on manual headcounts; subjective assumptions regarding product placement

		queue times, and monitor customer journey flow.	effectiveness; lack of empirical data to justify labor costs.
Security Personnel	Loss Prevention Officer	Detect immediate physical threats (e.g., weapons), monitor restricted zones (e.g., stockrooms), and respond to incidents in real-time.	Alarm fatigue from traditional motion sensors; inability to monitor multiple camera feeds simultaneously; delayed response times due to post-incident discovery.

3. User Stories & Use Cases

- **US-01 (Weapon Threat Alert):** As a *Small Business Owner*, I want to receive an instantaneous system alert (via dashboard and webhook) when a weapon is detected, so that I can ensure the immediate safety of my employees and customers.
- **US-02 (Spatial Analytics):** As a *Store Manager*, I want to view a dynamically generated heatmap of my store layout over a configurable timeframe, so that I can optimally position high-margin products in high-traffic zones.
- **US-03 (Foot Traffic Validation):** As a *Shift Supervisor*, I want to check accurate daily customer entry and exit counts via an analytics dashboard, so that I can validate foot traffic metrics against point-of-sale (POS) conversion rates.
- **US-04 (Restricted Zone Monitoring):** As a *Security Personnel*, I want to be automatically notified when an individual crosses a virtual tripwire into a restricted zone (e.g., after-hours stockroom), so that I can promptly investigate unauthorized access.
- **US-05 (Staffing Optimization):** As a *Store Manager*, I want to view time-based behavior analysis showing peak and off-peak hours, so that I can create data-driven staff schedules that reduce operational overhead without sacrificing customer service.

4. Functional Requirements

- **FR-01: Real-Time Person Detection:** The system shall process video streams to identify and localize “persons” using bounding boxes with a minimum confidence threshold of 70%, disregarding non-human moving objects (e.g., shopping carts, shadows).

- **FR-02: Unique ID Assignment (Tracking):** The system shall leverage Multi-Object Tracking (ByteTrack) to assign and maintain a unique numerical identifier (ID) to each detected person for the duration of their presence in the camera's Field of View (FOV).
 - **FR-03: Entry/Exit Line-Crossing Logic:** The system shall allow administrators to define virtual linear boundaries on the dashboard and accurately increment/decrement traffic counters based on the directional vector of tracked IDs crossing these lines.
 - **FR-04: Restricted Zone Intrusion Detection:** The system shall support the configuration of polygonal geographic zones; upon a tracked ID entering a designated "Restricted" zone, the system shall trigger an immediate intrusion event.
 - **FR-05: Weapon Threat Identification:** The system shall run a secondary, highly specific YOLOv8 model to detect predefined weapon classes (e.g., firearms, bladed weapons) and flag the associated tracked ID, triggering a critical security alert.
 - **FR-06: Real-Time Analytics Dashboard:** The system shall provide a Next.js-based web interface displaying live video feeds overlaid with bounding boxes/IDs, real-time numerical counters, active alert logs, and historical heatmap renderings.
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5. Non-Functional Requirements

- **NFR-01 (Performance):** The system shall maintain an end-to-end frame processing and API broadcast latency of **< 200 milliseconds** per frame to ensure perceived real-time synchronization between the physical environment and the web dashboard.
- **NFR-02 (Usability):** The frontend dashboard shall adhere to responsive web design principles, ensuring that critical live feeds and alert mechanisms are accessible and readable on both desktop monitors and mobile devices without requiring specialized training.
- **NFR-03 (Security & Privacy):** The system shall enforce a strict privacy-first architecture; it shall explicitly omit facial recognition, biometric extraction, and personal identity logging. Visual output shall be restricted to anonymized bounding boxes and skeletal/centroid tracking points to ensure compliance with global data protection standards.
- **NFR-04 (Reliability):** The backend microservices architecture shall implement automated health checks and error-recovery mechanisms, targeting a **99% uptime** availability metric during configured active business hours.
- **NFR-05 (Scalability):** The system architecture shall be designed using decoupled, asynchronous message brokering (via WebSockets) to horizontally scale and

support concurrent ingestion of multiple independent camera streams without linear degradation of the frame processing rate.